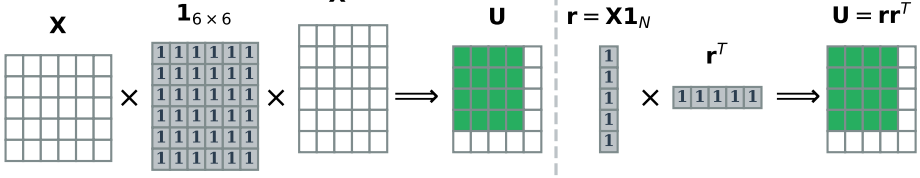
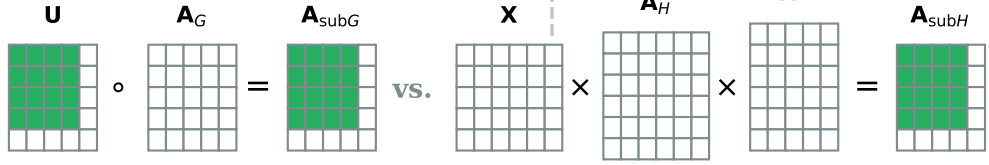
(a) Naive Evaluation: $\mathbf{U} = \mathbf{X}\mathbf{1}_{N\times N}\mathbf{X}^T \in \mathbb{R}^{M\times M}$ $[\Theta(MN^2 + M^2N)]$ **(b) Lemma 1 Rank-1 Factorization:** $\mathbf{U} = \mathbf{r}\mathbf{r}^T$ $[\Theta(MN + M^2)]$



(c) Subgraph Extraction: $\mathbf{A}_{\text{sub}G} = \mathbf{U} \circ \mathbf{A}_G, \quad \mathbf{A}_{\text{sub}H} = \mathbf{X}\mathbf{A}_H\mathbf{X}^T$



Objective Residual & Theorem 1 Exact Analytical Gradient:

Residual Matrix: $\mathbf{R} = \mathbf{A}_{\text{sub}G} - \mathbf{A}_{\text{sub}H} = (\mathbf{U} \circ \hat{\mathbf{A}}_G) - (\mathbf{X}\hat{\mathbf{A}}_H\mathbf{X}^T)$

$\nabla_{\mathbf{X}} F(\mathbf{X}) = 4(\mathbf{R} \circ \hat{\mathbf{A}}_G)\mathbf{r}\mathbf{1}_N^T - 4\mathbf{R}\mathbf{X}\hat{\mathbf{A}}_H \implies \Theta(N^3)$ **Closed-Form Evaluation (Zero Approximations)**